Politecnico di Torino

Linkable Ring Signatures from Group Actions

Constructions, Pitfalls, and New Approaches

Leonardo Errati

Chalmers S&P seminars, 29 Oct 2026

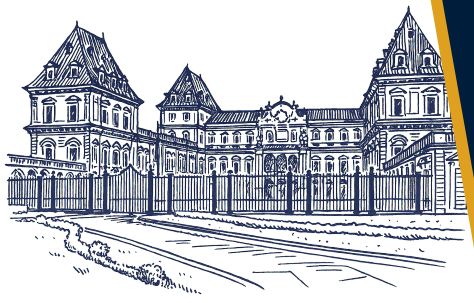


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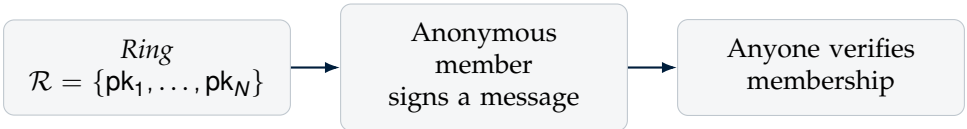
- Linkable ring signatures
- Group actions in cryptography
- LRS from group actions: BKP framework
- LRS from group actions: Canonical Lifting framework

So many signatures!

Primitive	What is established?	Motivation
Ordinary	This key authorised m.	Software updates, authentication...
Ring	One key in a public set authorised m.	Anonymous endorsement, whistleblowing...
Linkable Ring	As above, plus detection of repeated use of the same key.	Duplicate voting, double spending...
Group	Membership in a managed group; often authorised opening.	Anonymous membership with accountability.

Ring signatures

An organisation publishes a list of eligible public keys.



What we like:

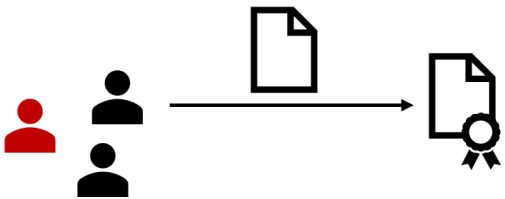
- to learn that an eligible key signed.

What we don't:

- to learn which key signed.
- attackers able to forge a signature.

Ring signatures

One signer and many public keys.



$(\mathsf{pk}_i, \mathsf{sk}_i) \leftarrow \mathsf{KeyGen}(\mathsf{pp})$

$\mathcal{R} = \{\mathsf{pk}_1, \ldots, \mathsf{pk}_N\}$

$\sigma \leftarrow \mathsf{Sign}(\mathsf{sk}_i, m, \mathcal{R})$

$b \leftarrow \mathsf{Verify}(\mathcal{R}, m, \sigma), \quad b \in \{0, 1\}$

Rivest, Shamir, and Tauman, "How to leak a secret", 2001

Ring signatures: trouble in the application layer

Yes,

Users generate their own keys.
Effective signer selects public keys.
Verification binds signature to message and ring.

but...

Who is eligible to register keys?
Is the ring fixed, are decoys sampled, etc?
How are keys authenticated, validated, revoked?

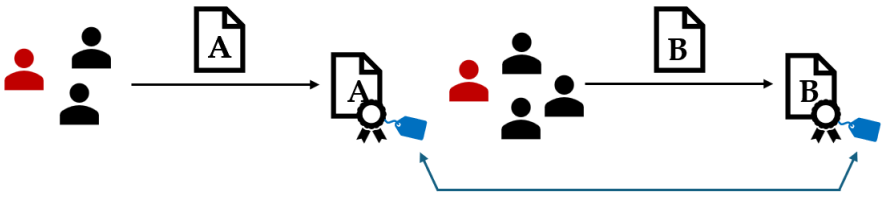
More troubles

A large ring is not automatically a large anonymity set (malicious keys, metadata, overlapping rings...).



Linkable ring signatures

Publicly recognise repeated use of a signing key without identifying it. Often implemented with a **linking tag** bound to the signing key.



$\sigma = (\sigma', \tau)$ $\qquad b \leftarrow \mathsf{Link}(\sigma_a, \sigma_b), \qquad b \in \{0, 1\}.$

A controlled loss of privacy able to reveal repetition.

Liu, Wei, and Wong, "Linkable spontaneous anonymous group signature for ad hoc groups", 2004

RS vs LRS

Application	Why anonymity?	RS or LRS?
Private payments	Hide which output is spent.	LRS: reject repeated spending.
Voting / survey	Hide the participating member.	LRS: one use per key and context.
Whistleblowing	Protect an insider's identity.	RS for unlinkability; LRS to recognise a returning source.

Key policy must be part of the application. Decide eligibility, ring selection, link scope, revocation...

Example: Monero's key images

A payment hides which output it spent among a ring of real plus decoy outputs.



Double spending repeats I, even with a different ring.

Key image I?

A public fingerprint derived from the spent output's secret key. Repeats on reuse.

koe, Alonso, and Noether, *Zero to Monero*, 2020

Example: anonymous voting



LRS does not provide ballot secrecy, coercion resistance, one-person-one-key registration, verifiable tally...

Which scope?

Persistent keys with global linking? Dedicated per-election keys?

Example: whistleblowing



Crypto-Dissent, anonymous chat

Authenticate members before joining a chat. Recognise repeat authentication by the same credential.

Crypto-Drop, anonymous sources

Authenticate a document as coming from an organisation's member. Journalists can link later submissions from the same source.

Metadata

LRS hide the signing key within the ring, not IP addresses or document metadata.

Maheswaran, Jackowitz, Wolinsky, Wang, and Ford, *Crypto-Book: Bootstrapping Privacy Preserving Online Identities from Social Networks*, 2014

Security properties: linkability

Idea: with N keys an attacker cannot create $N+1$ valid and pairwise unlinked signatures.

$\mathsf{Game}^{\mathsf{Link}}_{\mathcal{A}}(\lambda)$

1: $\mathsf{pp} \leftarrow \mathsf{Setup}(1^\lambda)$
2: $(PK, \{(\sigma_j, m_j, \mathcal{R}_j)\}_{j=1}^{N+1}) \leftarrow \mathcal{A}(\mathsf{pp})$, where $|PK| = N$
3: **for** $j = 1, \ldots, N+1$ **do**
4: **if** $\mathcal{R}_j \not\subseteq PK$ or $\mathsf{Verify}(\mathcal{R}_j, m_j, \sigma_j) \neq \mathsf{accept}$ **then**
5: **return** 0
6: **end if**
7: **end for**
8: **for** all $i < j$ **do**
9: **if** $\mathsf{Link}(\sigma_i, \sigma_j) \neq \mathsf{reject}$ **then return** 0
10: **end for**
11: **return** 1

Beullens, Katsumata, and Pintore, "Calamari and Falafl: logarithmic (linkable) ring signatures from isogenies and lattices", 2020

Security properties: linkable anonymity

Idea: link pattern must be visible but identities must be hidden.

$\mathsf{Game}_{\mathcal{A}}^{\text{L-Anon}}(\lambda)$

1: $\mathsf{pp} \leftarrow \mathsf{Setup}(1^\lambda)$
2: $(\mathsf{pk}_i, \mathsf{sk}_i) \leftarrow \mathsf{KeyGen}(\mathsf{pp}),\ i \in [N]$
3: $PK \leftarrow \{\mathsf{pk}_i : i \in [N]\};\ b \xleftarrow{\$} \{0,1\}$
4: $(\mathsf{pk}_0^*, \mathsf{pk}_1^*) \leftarrow \mathcal{A}(\mathsf{pp}, PK)$
5: Require distinct $\mathsf{pk}_0^*, \mathsf{pk}_1^* \in PK$
6: $K \leftarrow \{(\mathsf{pk}_i, \mathsf{sk}_i) :$
$\quad \mathsf{pk}_i \notin \{\mathsf{pk}_0^*, \mathsf{pk}_1^*\}\}$
7: $b^* \leftarrow \mathcal{A}^{\mathcal{O}_{\text{sign},b}}(K)$
8: **return** $[b^* = b]$

$\mathcal{O}_{\text{sign},b}(\mathsf{pk}, m, \mathcal{R})$

1: Require $\mathsf{pk} \in \{\mathsf{pk}_0^*, \mathsf{pk}_1^*\} \subseteq \mathcal{R}$
2: **if** $\mathsf{pk} = \mathsf{pk}_0^*$ **then**
3: $\quad$ **return** $\mathsf{Sign}(\mathsf{sk}_b^*, m, \mathcal{R})$
4: **else**
5: $\quad$ **return** $\mathsf{Sign}(\mathsf{sk}_{1-b}^*, m, \mathcal{R})$
6: **end if**

For input pk_j^*, channel j signs with $\mathsf{sk}_{j \oplus b}^*$.
Non-challenge keys in $\mathcal{R}$ may be adversarial.

Beullens, Katsumata, and Pintore, "Calamari and Falafl: logarithmic (linkable) ring signatures from isogenies and lattices", 2020

Security properties: one-time linkable anonymity

Idea: link pattern must be visible but identities must be hidden **under one signature**.

$\mathsf{Game}_{\mathcal{A}}^{1\text{-Anon}}(\lambda)$

$\mathsf{pp} \leftarrow \mathsf{Setup}(1^\lambda)$
$(\mathsf{pk}_i, \mathsf{sk}_i) \leftarrow \mathsf{KeyGen}(\mathsf{pp}),\ i \in [N]$
$PK \leftarrow \{\mathsf{pk}_i : i \in [N]\};\ b \xleftarrow{\$} \{0,1\}$
$(\mathsf{pk}_0^*, \mathsf{pk}_1^*) \leftarrow \mathcal{A}(\mathsf{pp}, PK)$
Require distinct $\mathsf{pk}_0^*, \mathsf{pk}_1^* \in PK$
$K \leftarrow \{(\mathsf{pk}_i, \mathsf{sk}_i) : \mathsf{pk}_i \notin \{\mathsf{pk}_0^*, \mathsf{pk}_1^*\}\}$
$\mathsf{used} \leftarrow 0$
$b^* \leftarrow \mathcal{A}^{\mathcal{O}^{(1)}_{\mathrm{sign},b}}(K)$
return $[b^* = b]$

$\mathcal{O}^{(1)}_{\mathrm{sign},b}(\mathsf{pk}, m, \mathcal{R})$

Require $\mathsf{pk} \in \{\mathsf{pk}_0^*, \mathsf{pk}_1^*\} \subseteq \mathcal{R}$
if $\mathsf{used} = 1$ **then return** $\perp$
$\mathsf{used} \leftarrow 1$
if $\mathsf{pk} = \mathsf{pk}_0^*$ **then**
    return $\mathsf{Sign}(\mathsf{sk}_b^*, m, \mathcal{R})$
else
    return $\mathsf{Sign}(\mathsf{sk}_{1-b}^*, m, \mathcal{R})$
end if

Beullens, Katsumata, and Pintore, "Calamari and Falafl: logarithmic (linkable) ring signatures from isogenies and lattices", 2020
Bultel and Olivier-Anclin, "On the Anonymity of Linkable Ring Signatures", 2024

Security properties: non-frameability

An attacker cannot create a fresh valid signature that links to an honest user's signature.

$\mathsf{Game}_{\mathcal{A}}^{\text{Non-Frame}}(\lambda)$

1: $\mathsf{pp} \leftarrow \mathsf{Setup}(1^\lambda)$
2: $(\mathsf{pk}_i, \mathsf{sk}_i) \leftarrow \mathsf{KeyGen}(\mathsf{pp}),\ i \in [N]$
3: $PK \leftarrow \{\mathsf{pk}_i : i \in [N]\}$
4: $SL, CL, T \leftarrow \emptyset$
5: $(\mathcal{R}^*, m^*, \sigma^*) \leftarrow \mathcal{A}^{\mathcal{O}_{\text{sign}}, \mathcal{O}_{\text{corrupt}}}(\mathsf{pp}, PK)$
6: **return** the conjunction:
$\mathsf{Verify}(\mathcal{R}^*, m^*, \sigma^*) = \mathsf{accept}$
$(\cdot, m^*, \mathcal{R}^*) \notin SL$
$\exists (i, \sigma) \in T :\ \mathsf{pk}_i \notin CL \wedge \mathsf{Link}(\sigma^*, \sigma) = \mathsf{accept}$

$\mathcal{O}_{\text{sign}}(i, m, \mathcal{R})$

1: Require $i \in [N]$ and $\mathsf{pk}_i \in \mathcal{R}$
2: $\sigma \leftarrow \mathsf{Sign}(\mathsf{sk}_i, m, \mathcal{R})$
3: $SL \leftarrow SL \cup \{(i, m, \mathcal{R})\}$
4: $T \leftarrow T \cup \{(i, \sigma)\}$
5: **return** σ

$\mathcal{O}_{\text{corrupt}}(i)$

1: Require $i \in [N]$
2: $CL \leftarrow CL \cup \{\mathsf{pk}_i\}$
3: **return** $(\mathsf{pk}_i, \mathsf{sk}_i)$

SL: signing queries $(i, m, \mathcal{R})$.
T: returned signatures (i, σ). CL: corrupted public keys.

Beullens, Katsumata, and Pintore, "Calamari and Falafl: logarithmic (linkable) ring signatures from isogenies and lattices", 2020

Security properties: unforgeability

An attacker cannot sign for a ring consisting entirely of uncorrupted users.

$\mathsf{Game}^{\mathsf{Unf}}_{\mathcal{A}}(\lambda)$

1: $\mathsf{pp} \leftarrow \mathsf{Setup}(1^\lambda)$
2: $(\mathsf{pk}_i, \mathsf{sk}_i) \leftarrow \mathsf{KeyGen}(\mathsf{pp}),\ i \in [N]$
3: $PK \leftarrow \{\mathsf{pk}_i : i \in [N]\};\ SL, CL \leftarrow \emptyset$
4: $(\mathcal{R}^*, m^*, \sigma^*) \leftarrow \mathcal{A}^{\mathcal{O}_{\mathrm{sign}}, \mathcal{O}_{\mathrm{corrupt}}}(\mathsf{pp}, PK)$
5: **return** the conjunction:
$\mathcal{R}^* \subseteq PK \setminus CL$
$(\cdot, m^*, \mathcal{R}^*) \notin SL$
$\mathsf{Verify}(\mathcal{R}^*, m^*, \sigma^*) = \mathsf{accept}$

(oracles and logs as before)

Implied by the other properties

Compare a forgery with one fresh honest signature per ring key.

- Links to an honest signature?
  Non-frameability is broken.
- No such link?
  Linkability is broken.

Beullens, Katsumata, and Pintore, "Calamari and Falafl: logarithmic (linkable) ring signatures from isogenies and lattices", 2020

Take-aways

1. Only one member actively signs.
2. In LRS, we also reveal repeated use.
3. So many application-layer decisions! Eligibility, ring policy, and so on.

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Why post-quantum?

Assumption diversity:
quantum attacks (Shor, Grover) threaten past primitives.

Functional diversity:
post-quantum ordinary signature does not automatically get advanced functionalities.



NIST, *Additional Digital Signature Schemes*, 2026

Transition now

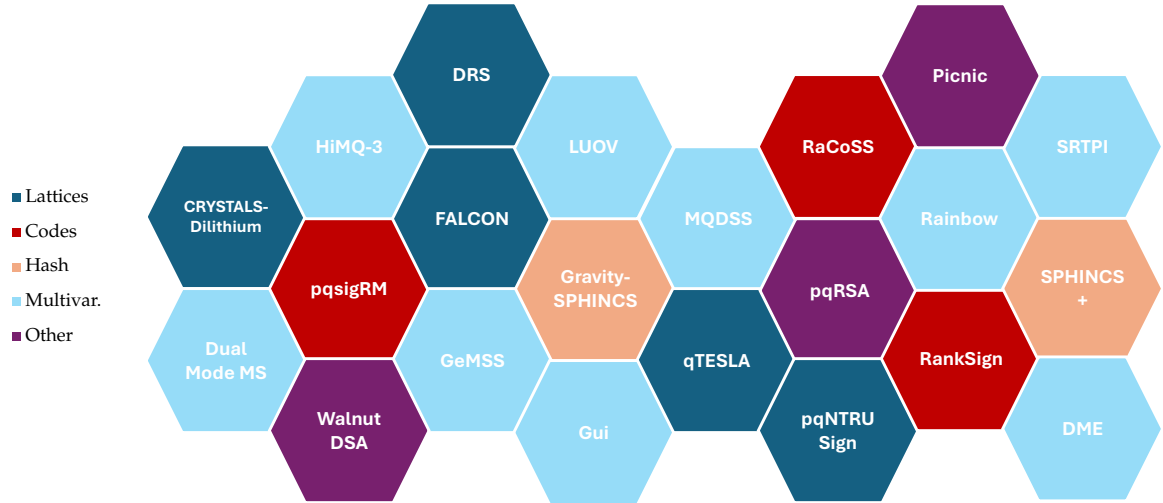
NIST IR 8547 (October 2026): transition plan for digital signatures.

Digital Signature Algorithm Family	Parameters	Transition
ECDSA [FIPS186]	≥ 112 bits of classical security strength	***Deprecated*** after 2030 ***Disallowed*** after 2035
EdDSA [FIPS186]	≥ 128 bits of classical security strength	***Deprecated*** after 2030 ***Disallowed*** after 2035
RSA [FIPS186]	≥ 112 bits of classical security strength	***Deprecated*** after 2030 ***Disallowed*** after 2035

Moody, Perlner, Regenscheid, Robinson, and Cooper, *Transition to Post-Quantum Cryptography Standards*, 2026, Table 2 and Secs. 4.1–4.1.1, pp. 14–15

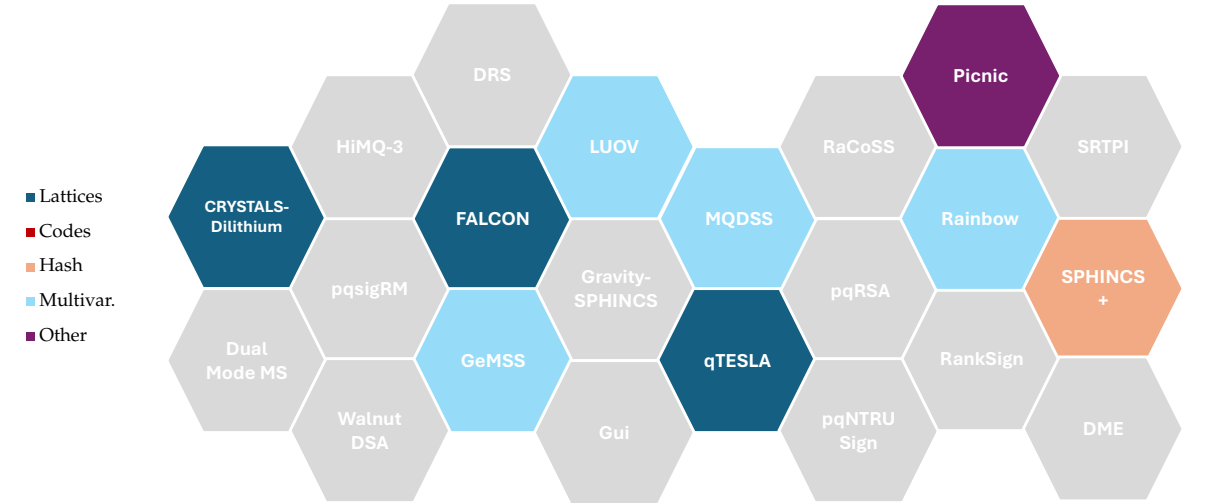
Original NIST selection

2017–2022



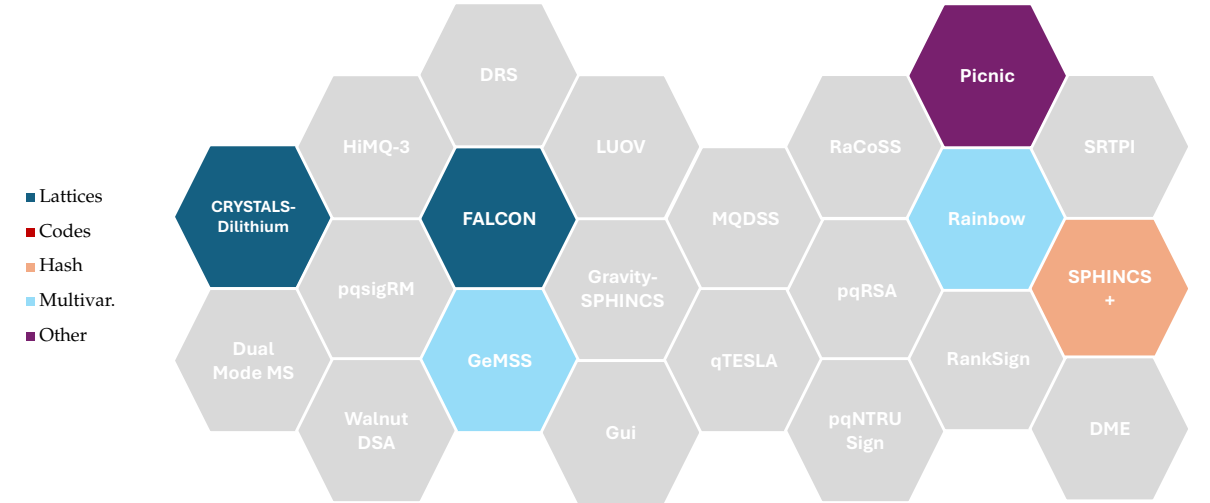
Original NIST selection

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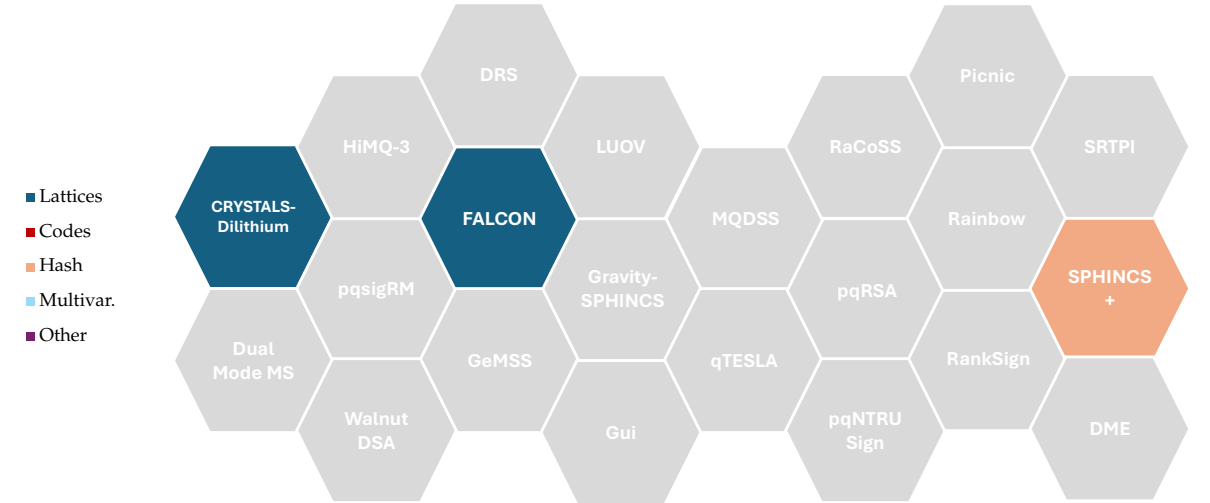
Original NIST selection

2017–2022



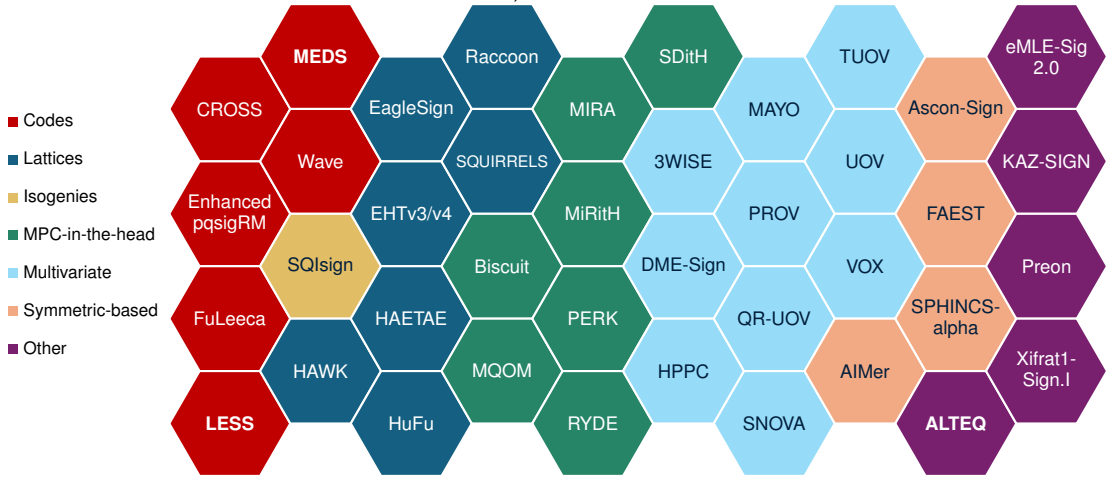
Original NIST selection

2017–2022



NIST Additional Digital Signatures

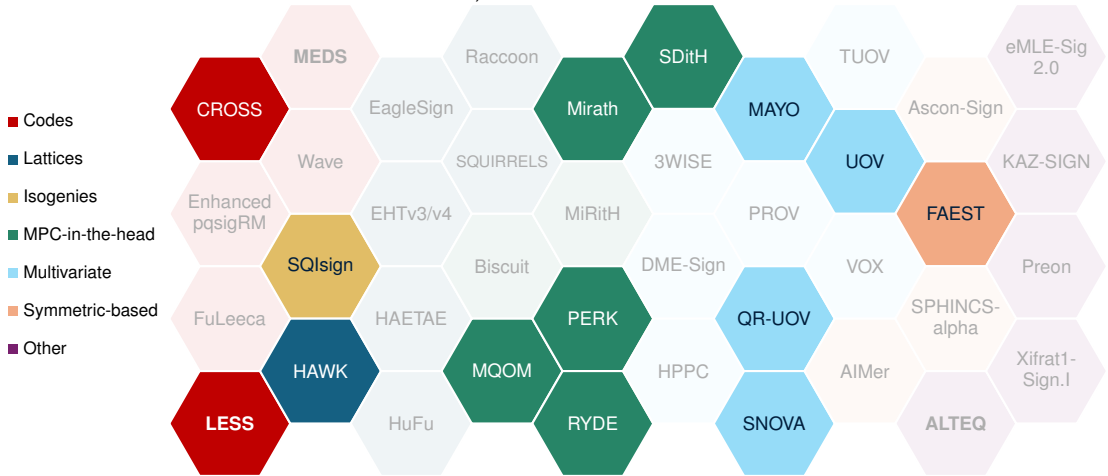
Round 1, 2023: 40 candidates



NIST, *Additional Digital Signature Schemes*, 2026; NIST, *Nine Candidates Advance to the Third Round of the Additional Digital Signatures*, 2026.

NIST Additional Digital Signatures

Round 2, October 2024: 14 candidates

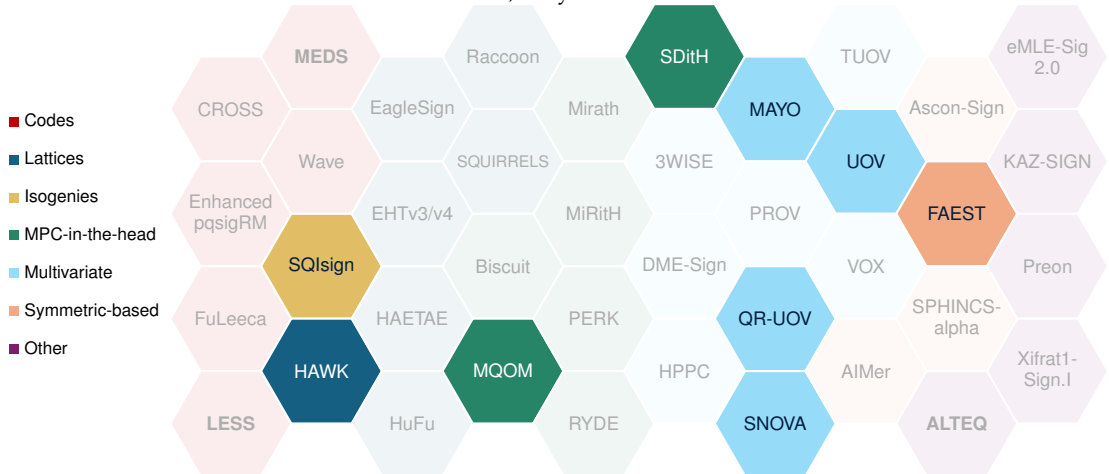


MIRA and MiRitH merged in Mirath.

NIST, *Additional Digital Signature Schemes*, 2026; NIST, *Nine Candidates Advance to the Third Round of the Additional Digital Signatures*, 2026.

NIST Additional Digital Signatures

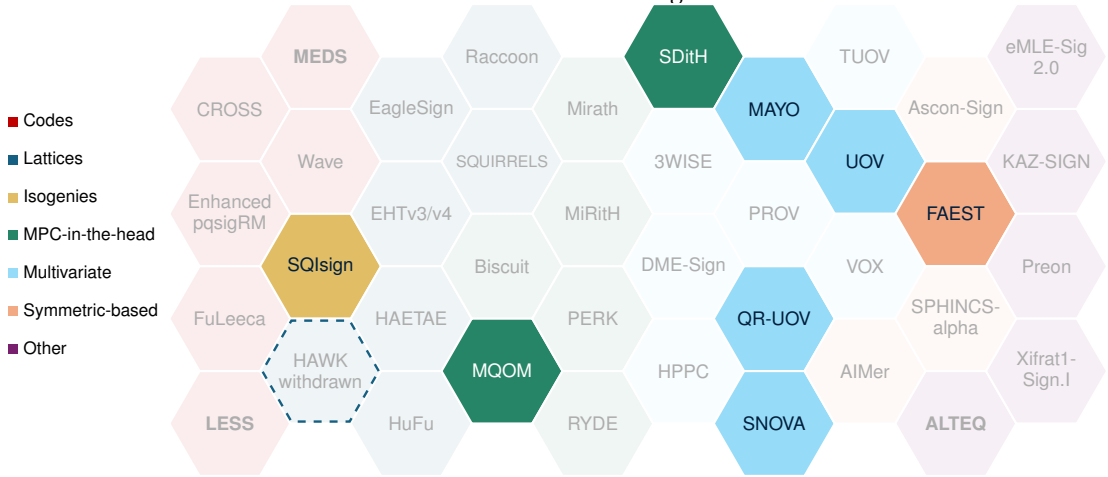
Round 3, May 2026: 9 candidates



NIST, *Additional Digital Signature Schemes*, 2026; NIST, *Nine Candidates Advance to the Third Round of the Additional Digital Signatures*, 2026.

NIST Additional Digital Signatures

October 2026: 8 remaining candidates



HAWK withdrawn, 29 July 2026.

NIST, *Additional Digital Signature Schemes*, 2026; NIST, *Nine Candidates Advance to the Third Round of the Additional Digital Signatures*, 2026.

Group actions



$(G, \cdot)$: transformations, with composition.
X: objects on which the transformations act.

Definition: a left group action

A map $\star : G \times X \to X$ such that

$$e \star x = x, \qquad (gh) \star x = g \star (h \star x).$$

Commutativity is not required. For Rubik's cube,

$$X = \{\text{possible states of the cube}\},$$
$$G \cong (\mathbb{Z}_3^7 \times \mathbb{Z}_2^{11}) \rtimes ((A_8 \times A_{12}) \rtimes \mathbb{Z}_2) \leq S_{48}.$$

Group actions in cryptography

Recovering a secret transformation from input and output should be hard.

$$\underbrace{x}_{\text{public base}} \xrightarrow{g} \underbrace{y = g \star x}_{\text{public key}}.$$

Group Action Inverse Problem

Given (x, y), find some g with $y = g \star x$. The witness need not be unique.

Some instantiations are post-quantum!

Objects	Transformations	Hard problem
Linear codes	Coordinate permutations and scalings	Code equivalence
Lattices	Integral changes of basis	Lattice isomorphism

Example: code equivalence

Objects: k-dimensional linear codes $C = \mathsf{rowspan}(G) \subseteq \mathbb{F}_q^n$.
Transformations: coordinate permutations P and non-zero scalings D.

$$G' = SGPD, \qquad S \in \mathrm{GL}_k(\mathbb{F}_q), \qquad C' = C \cdot (PD).$$

$$\underset{G}{\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}} \xrightarrow{PD} \underset{GPD}{\begin{pmatrix} 1 & 0 & 1 \\ 1 & 2 & 0 \end{pmatrix}} \xrightarrow{S} \underset{G' = SGPD}{\begin{pmatrix} 2 & 2 & 1 \\ 1 & 2 & 0 \end{pmatrix}}$$

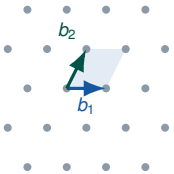
Hard problem: recover a coordinate transformation.

Biasse, Micheli, Persichetti, and Santini, "LESS is More: Code-Based Signatures Without Syndromes", 2020

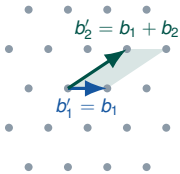
Example: lattice isomorphism

Objects: Gram matrices $Q = B^\top B$ (basis lengths & angles).
Transformations: unimodular matrices $U \in \mathrm{GL}_n(\mathbb{Z})$, i.e. $\det U = \pm 1$.

$$U = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right), \qquad B' = BU, \qquad Q' = Q \cdot U = U^\top Q U \quad \text{(right action)}.$$



$$B = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}, \quad Q = \begin{pmatrix} 4 & 2 \\ 2 & 5 \end{pmatrix}.$$



$$B' = \begin{pmatrix} 2 & 3 \\ 0 & 2 \end{pmatrix}, \quad Q' = \begin{pmatrix} 4 & 6 \\ 6 & 13 \end{pmatrix}.$$

Hard problem: given Q, Q', find a suitable U.

Budroni, Chi-Domínguez, and Franch, "Don't Use It Twice: Reloaded! On the Lattice Isomorphism Group Action", 2025

Take-aways

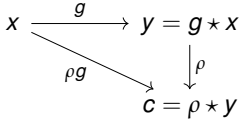
1. A group describes transformations, the action describes their effect.
2. Must be efficient to compute but with hard computational problems.
3. Codes and lattices provide candidate hard problems.

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Base sigma protocol

Prover knows g such that $y = g \star x$.



$$\mathcal{R}_{\mathrm{GA}} = \{((x, y), g) : y = g \star x\}.$$

Challenge 0
reveal ρ; check $c = \rho \star y$

Challenge 1
reveal ρg; check $c = (\rho g) \star x$

OR sigma protocol

Fix the public base x and ring $(y_1, \ldots, y_N)$.

$$\mathcal{R}_{\mathrm{GA}}^{\mathrm{OR}} = \{((x, y_1, \ldots, y_N), (i, g)) : i \in [N] \ \wedge \ y_i = g \star x\}.$$

1. Randomise keys applying the same mask ρ, then commit with independent r_j:

$$a_j = \rho \star y_j, \qquad c_j = \mathsf{Com}(a_j; r_j), \qquad j \in [N].$$

2. **Index-hiding Merkle tree:** leaves $H(c_j)$, sorted child hashes u, v: $H\big(\min(u, v) \parallel \max(u, v)\big)$.

Challenge 0: reconstruct every branch
Reveal the seed generating ρ and all r_j.
Recompute all c_j from the public ring, check the root.

Challenge 1: open the known branch
Reveal $z = \rho g$, r_i, and an index-hiding path.
Reconstruct $\mathsf{Com}(z \star x; r_i)$, check its path against R.

Beullens, Katsumata, and Pintore, "Calamari and Falafl: logarithmic (linkable) ring signatures from isogenies and lattices", 2020, Uniform-group illustration.

BKP framework: OR sigma protocol with tag

Given two actions

$$\star : G \times X \to X, \qquad \bullet : G \times T \to T.$$

prove that the same witness generates one public key and the tag:

$$\mathcal{R}_{\mathrm{GA}}^{\text{OR-t}} = \{((x, y_1, \ldots, y_N, t, \tau), (i, g)) : y_i = g \star x \ \wedge \ \tau = g \bullet t\}.$$

Parallel repetitions + Fiat-Shamir = LRS.
Link using a tag relation (equality, closeness...).

Beullens, Katsumata, and Pintore, “Calamari and Falafl: logarithmic (linkable) ring signatures from isogenies and lattices”, 2020

BKP framework: security properties

$$\underbrace{y = g \star x}_{\text{public key}}, \qquad \underbrace{\tau = g \bullet t}_{\text{linking tag}}, \qquad \underbrace{z = \rho g}_{\text{masked proof response}} .$$

Desired guarantee	What the pair of actions must prevent
Linkability	Two witnesses for one public key yielding unlinked tags.
Linkable anonymity	Recognising which public key is correlated with a visible tag.
Non-frameability	Finding a new witness whose tag links to an honest user's tag.

Beullens, Katsumata, and Pintore, "Calamari and Falafl: logarithmic (linkable) ring signatures from isogenies and lattices", 2020

BKP framework: admissible pairs

$$\underbrace{y = g \star x}_{\text{public key}}, \qquad \underbrace{\tau = g \bullet t}_{\text{linking tag}}, \qquad \underbrace{z = \rho g}_{\text{masked proof response}} .$$

1. Efficient algorithms for the actions, sampling, evaluation, encoding.
2. Efficient algorithm for evaluating $\mathsf{Link}_{\mathrm{GA}}(\tau, \tau')$, with $\mathsf{Link}_{\mathrm{GA}}(\tau, \tau) = 1$.
3. Accepted responses lie in a common, efficiently recognisable domain and have a distribution independent of the secret g.

More related to security properties:

4. The public pair must look like independent uniform elements of $X \times T$.
5. Two valid witnesses for the same public key must not yield unlinked tags.
6. Finding a new witness whose tag links to it must be hard.

Beullens, Katsumata, and Pintore, "Calamari and Falafl: logarithmic (linkable) ring signatures from isogenies and lattices", 2020

Calamari and Falafl

Two instantiations:

Calamari	Falafl
Isogenies (CSIDH).	**Module lattices (MLWE/MSIS).**

Beullens, Katsumata, and Pintore, “Calamari and Falafl: logarithmic (linkable) ring signatures from isogenies and lattices”, 2020

Attack 1: Don't Use It Twice!

Reusing the **same secret transformation** can expose more information.
We have two views of the same hidden relabelling. One instance hides a permutation and coordinate scalings. A second instance adds constraints!

$$C_1 \xrightarrow{Q} C_1' = C_1 Q$$

$$C_2 \xrightarrow{Q} C_2' = C_2 Q$$

A candidate transformation must fit both pairs. The second pair can eliminate the ambiguity protecting the first.

Budroni, Chi-Domínguez, D'Alconzo, Di Scala, and Kulkarni, "Don't use it twice! Solving relaxed linear equivalence problems", 2024

Attack 1: Don't Use It Twice! (ctd)

$$C_1' = C_1 Q, \quad C_2' = C_2 Q \quad \Longrightarrow \quad G_j Q (H_j')^\top = 0.$$

Public inputs: G_j generates C_j; compute a parity-check matrix H_j' for C_j', so $G_j'(H_j')^\top = 0$.

$$\begin{pmatrix} M_1 = H_1' \otimes G_1 \\ M_2 = H_2' \otimes G_2 \end{pmatrix} \mathrm{vec}(Q) = 0$$

Stack both linear systems.
The same secret Q must satisfy both.

Guess one nonzero position of the *monomial* Q, every other entry in that row and column must be zero. At rate $k/n = 1/2$, the paper gives a heuristic polynomial-time recovery attack, supported by experiments.

Budroni, Chi-Domínguez, D'Alconzo, Di Scala, and Kulkarni, "Don't use it twice! Solving relaxed linear equivalence problems", 2024

Attack 2: Don't Use It Twice! Reloaded

For a public Gram matrix Q_0, the proposal uses a secret unimodular U:

$$Y = U^\top Q_0 U \quad \text{public key}, \qquad T = U^{-\top} Q_0 U^{-1} \quad \text{tag}.$$

Rearranging the tag equation gives $Q_0 = U^\top T U$. The key and tag therefore provide **two public pairs with the same secret**:

$$Q_0 \xrightarrow{\quad U \quad} Y$$

$$T \xrightarrow[\text{same } U]{\qquad\qquad} Q_0$$

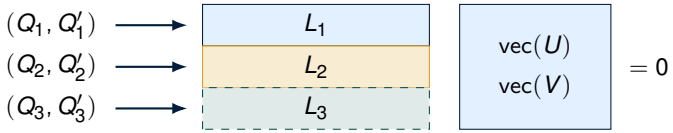
Testing whether a candidate key and the tag share this pattern can identify the signer.

Budroni, Chi-Domínguez, and Franch, "Don't Use It Twice: Reloaded! On the Lattice Isomorphism Group Action", 2025

Attack 2: Don't Use It Twice! Reloaded

$$Q'_j = U^\top Q_j U \implies Q'_j V - U^\top Q_j = 0, \qquad V = U^{-1}.$$

Temporarily treat U, V as independent unknowns.



2 instances: distinguish.
The rank detects enough information on the shared secret to break linkable anonymity.

3 instances: recover.
The kernel constrains (U, V), enforce $V = U^{-1}$ to recover $\pm U$.

Both results rely on heuristic rank assumptions, supported by experiments (Assumptions 1 and 2 in the paper).

Budroni, Chi-Domínguez, and Franch, "Don't Use It Twice: Reloaded! On the Lattice Isomorphism Group Action", 2025

Attack 3: non-commutativity

Correctness

One response Z must work for both

$$\text{key check } Z = UR,$$
$$\text{tag check } Z = RU.$$

An honest signature can be rejected.

Linkability

Different secret matrices can describe the same public key. The signer might get one public key with different tags. The same key can evade linking.

Iribarnegaray, Gregor, and Leal, *A note on the Unsuitability of LIGA for Linkable Ring Signatures: The perils of non-commutativity*, 2026

Take-aways: BKP

1. The same secret generates one public key in the ring and the linking tag. They must be secure together.
2. LRS security requires separate assumptions.

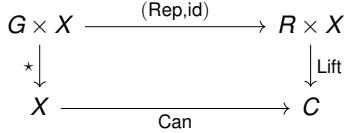
Table of Contents

- Linkable ring signatures
- Group actions in cryptography
- LRS from group actions: BKP framework
- LRS from group actions: Canonical Lifting framework

Canonical liftings

For a group action $(G, X, \star)$, take efficient maps

$$\mathsf{Rep} : G \to R, \qquad \mathsf{Can} : X \to C, \qquad \mathsf{Lift} : R \times X \to C.$$



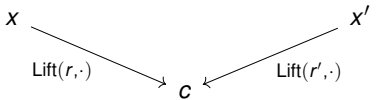
$$\mathsf{Can}(g \star x) = \mathsf{Lift}(\mathsf{Rep}(g), x).$$

Budroni, D'Alconzo, Errati, and Persichetti, *On Canonical Liftings, Group Actions, and Linkable Ring Signatures*, 2026

Canonical lifting collision problem (CLCP)

Given two public objects $x, x' \in X$, find representatives $r, r' \in R$ such that

$$\mathsf{Lift}(r, x) = \mathsf{Lift}(r', x').$$



A canonical lifting is **effective** if we have efficient algorithms for

1. Given any $r \in R$, find $g \in G$ such that $\mathsf{Rep}(g) = r$.
2. Given $u, v \in X$ with $\mathsf{Can}(u) = \mathsf{Can}(v)$, find $h \in G$ such that $v = h \star u$, if any.

For an effective lifting, CLCP and GAIP are polynomial-time equivalent.

Budroni, D'Alconzo, Errati, and Persichetti, *On Canonical Liftings, Group Actions, and Linkable Ring Signatures*, 2026

Canonical lifting for LRS

Same full witness

$$y = g \star x, \qquad \tau = g \bullet t.$$

Just same representative

$$y = g \star x, \quad \tau = \widetilde{g} \star t,$$
$$\mathsf{Rep}(g) = \mathsf{Rep}(\widetilde{g}).$$

Canonical lifting lets the verifier check this shared representative.

Budroni, D'Alconzo, Errati, and Persichetti, *On Canonical Liftings, Group Actions, and Linkable Ring Signatures*, 2026

Canonical lifting for LRS

Suppose $y = g \star x$, $\tau = \widetilde{g} \star t$, and $\mathsf{Rep}(g) = \mathsf{Rep}(\widetilde{g})$. A canonical lifting is **left-compatible** if:

1. For all $g, \widetilde{g}, \rho \in G$,
$$\mathsf{Rep}(g) = \mathsf{Rep}(\widetilde{g}) \quad \Longrightarrow \quad \mathsf{Rep}(\rho g) = \mathsf{Rep}(\rho \widetilde{g})$$
2. Given $g \in G$, we can efficiently sample uniformly from $[g]_{\mathsf{Rep}} = \{h \in G : \mathsf{Rep}(h) = \mathsf{Rep}(g)\}$.

Then one response $r = \mathsf{Rep}(\rho g) = \mathsf{Rep}(\rho \widetilde{g})$ checks both:



$$\mathsf{Lift}(r, x) = \mathsf{Can}(\rho \star y), \quad \mathsf{Lift}(r, t) = \mathsf{Can}(\rho \star \tau).$$

Budroni, D'Alconzo, Errati, and Persichetti, *On Canonical Liftings, Group Actions, and Linkable Ring Signatures*, 2026

$\mathsf{Rep}(S, PD) = P$	discard S and D
$\mathsf{Can}(SAD) = \mathsf{Can}(A)$	Use the canonical form of Duong et al.
$\mathsf{Lift}(P, A) = \mathsf{Can}(AP)$	Permute the columns of A by P, then apply Can.

Public key and linking tag

Fix two public generator matrices $G_0, T_0 \in \mathbb{F}_q^{k \times n}$. Choose secret P and independent $D, \widetilde{D}$:

$$\text{public key: } y = \mathsf{RREF}(G_0 PD), \qquad \text{tag: } \tau = \mathsf{RREF}(T_0 P \widetilde{D}).$$

Link signatures by comparing their canonical tags:

$$\mathsf{Link}(\sigma, \sigma') = 1 \iff \mathsf{Can}(\tau) = \mathsf{Can}(\tau').$$

Budroni, D'Alconzo, Errati, and Persichetti, *On Canonical Liftings, Group Actions, and Linkable Ring Signatures*, 2026
Duong, Khuc, Qiao, Susilo, and Zhang, "(Partially) Blind Signatures from Cryptographic Group Actions", 2026

Security assumptions

Assumption	Intended meaning
Non-splitting	A key cannot support incompatible canonical tags.
2-pseudorandomness	A shared representative is hidden in the public key/tag pair.
Frame one-wayness	An honest user's canonical tag cannot be forged into a fresh signature.

Budroni, D'Alconzo, Errati, and Persichetti, *On Canonical Liftings, Group Actions, and Linkable Ring Signatures*, 2026

Take-aways: canonical liftings

1. Key and tag can use different secrets. They share $\mathsf{Rep}(g) = \mathsf{Rep}(\widetilde{g})$, left compatibility preserves this under masking.
2. Effectiveness connects the new search problem to GAIP.
3. LRS security requires separate assumptions.

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